

QIAN TANG

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RESEARCH FOCUS

My research interests lie at the intersection of machine learning, optimization, and statistical computation, with a particular focus on the development of efficient algorithms for analyzing modern and complex data structure.

ACADEMIC EXPERIENCE

University of Minnesota IRSA Faragher Distinguished Postdoctoral Fellow	Minneapolis, MN Aug 2025 - Now
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EDUCATION

University of Iowa Ph.D. in Statistics	Iowa City, IA Aug 2020 - Aug 2025
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- Thesis: Topics in Statistical Learning Methods and Algorithms
- Thesis advisor: Boxiang Wang

University of California, Davis M.S. in Statistics	Davis, CA 2017 - 2019
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Jilin University B.S. in Information and Computation Science	Changchun, China 2012 - 2016
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AWARDS

IMS New Researchers Travel Award	2026
NeurIPS 2025 Scholar Award	2025
Hogg and Craig Award, University of Iowa	2024
Graduate College Summer Fellowship, University of Iowa	2023
Travel Award at the Statistics Annual Winter Workshop, University of Florida	2023

PUBLICATION

- Q. Tang, Y. Gu, and B. Wang. [QuanDA: Quantile-Based Discriminant Analysis for High-Dimensional Imbalanced Classification](#). In *The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025
- Q. Tang, Y. Gu, and B. Wang. [fastkqr: A Fast Algorithm for Kernel Quantile Regression](#). *Journal of Computational and Graphical Statistics*, 0(0):1–11, 2025
- Q. Tang, Y. Zhang, and B. Wang. [Finite Smoothing Algorithm for High-Dimensional Support Vector Machines and Quantile Regression](#). In *Forty-first International Conference on Machine Learning (ICML)*, 2024
- Q. Tang, E. Chi, and W. Wang. The ATLAS Penalty: Auxiliary-Transformed Location-Aware Smoothing with Applications to Spatial Transcriptomics. In *The STAI-X Conference*, 2026

PAPERS UNDER REVIEW

Q. Tang, X. Cai, and Y.-F. Chen. Weighted win score ratio for composite endpoints with continuous components. 2026. Submitted to Journal of Biopharmaceutical Statistics

IN PREPARATION

Q. Tang, E. Chi, and W. Wang. The ATLAS Penalty: Auxiliary-Transformed Location-Aware Smoothing with Applications to Spatial Transcriptomics. In *The STAI-X Conference*, 2026

Q. Tang and G. Jones. Bayesian quantile-based discriminant analysis. 2025+. In preparation

Q. Tang and G. Jones. Bayesian quantile regression with a generalized bridge prior. 2025+. In preparation

Q. Tang, B. Wang, W. Zhou, and H. Zou. Transfer Learning for High-Dimensional Clustering Analysis. 2025+. In preparation

Q. Tang, Z. Tong, B. Wang, and Y. Zhang. A Novel SAFE Model for Predicting Climate-Related Extreme Losses. 2025+. In preparation

EXPERIENCE

Food and Drug Administration , ORISE Fellow	May 2024 - Aug. 2024
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Novartis , Biostatistician Intern	May 2025 - Aug. 2025
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INVITED TALKS

fastkqr: A fast algorithm for kernel quantile regression

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| • Machine Learning Seminar
Xidian University | Dec. 2024 |
| • Data Mining Group Seminar
University of Iowa | Mar. 2024 |
| • The 2023 Statistics Annual Winter Workshop
University of Florida | Jan. 2023 |
| • Statistics Student Organization Seminar
University of Iowa | Nov. 2023 |

Quantile-Based Discriminant Analysis for High-Dimensional Imbalanced Classification

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| • The Thirty-Ninth Annual Conference on Neural Information Processing Systems
San Diego, CA, USA | Dec. 2025 |
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Two Stories in Quantile Regression: Fast Computation and Imbalanced Classification

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| • The First International New Researchers Conference
Seville, Spain | Dec. 2025 |
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SOFTWARE

- [QuanDA](#), an R package for binary classification under high-dimensional settings
- [fastkqr](#), an R package for computing regularized paths for kernel quantile regression model
- [hdqr](#), an R package for fitting elastic-net-penalized quantile regression model
- [hdsvm](#), an R package for computing the solution paths for elastic-net-penalized support vector machine

- [SAFEPC](#), an R package for predicting climate-related extreme losses by fitting a frequency-severity model

TEACHING

Instructor at University of Minnesota:

- STAT 5021: Statistical Analysis (Spring 2026)

Teaching Assistant at University of Iowa for the following courses:

- STAT 1030: Statistics for Business (Fall 2021, Spring 2024)
- STAT 3510: Biostatistics (Spring 2022, Fall 2022, Spring 2023)
- STAT 1020: Elementary Statistics and Inference (Fall 2024)

SERVICE

Organizer

- UMN School of Statistics Seminar Series, Spring 2026

Reviewer: reviewed at least one manuscript for the following journals

- IEEE Transactions on Information Theory
- Journal of Biopharmaceutical Statistics
- Statistica Sinica

SKILLS

- **Programming Languages:** R, Python, Fortran, C
- **Language:** English (bilingual and biliterate), Chinese (native)

PROFESSIONAL MEMBERSHIPS

- Member, American Statistical Association